

CSCI 423 Fall 2019

HW1

Due Wednesday Sept 18th

Part one:

Solve the problems in the text book chapter one:

Exercise 1.3.5 (page 17) : problem number 7

Exercise 1.4.2 (page 22): problems: number 1 and 2.

Part two:

Consider the following algorithm:

Setup();

For($i=0; i \leq n ; i++$)

 Process();

Give the Big-Oh time complexity of the algorithm in terms of n for each of the following cases:

a) Setup()= $O(1)$

process()= $O(1)$

b) Setup()= $O(\log n)$

process ()= $O(1)$

c) Setup ()= $O(n^2)$

Process()= $O(n^2)$

d) Setup()= $O(n^3)$

process()= $O(2^n)$